

FIIT STU - CLOUD COMPUTING 2026

Automated OpenStack Cinder Backup Management

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A GUI-Driven Approach using Python SDK & Docker

The Problem & Objective

- Manual management in OpenStack Horizon is time-consuming and error-prone.
- Backup "lifecycle" is often ignored, leading to uncontrolled storage costs.
- **Goal:** Automate the discovery and snapshot creation of *'in-use'* volumes.



-90% Time

Reduction in manual administration tasks.



100% Coverage

Ensuring all active volumes are protected.

System Architecture

Streamlit Frontend → Python SDK
→ OpenStack Cinder

GUI Layer

Logic Layer

Cloud Layer

Stateless Design: The application interacts with the Cloud API without local database dependency.

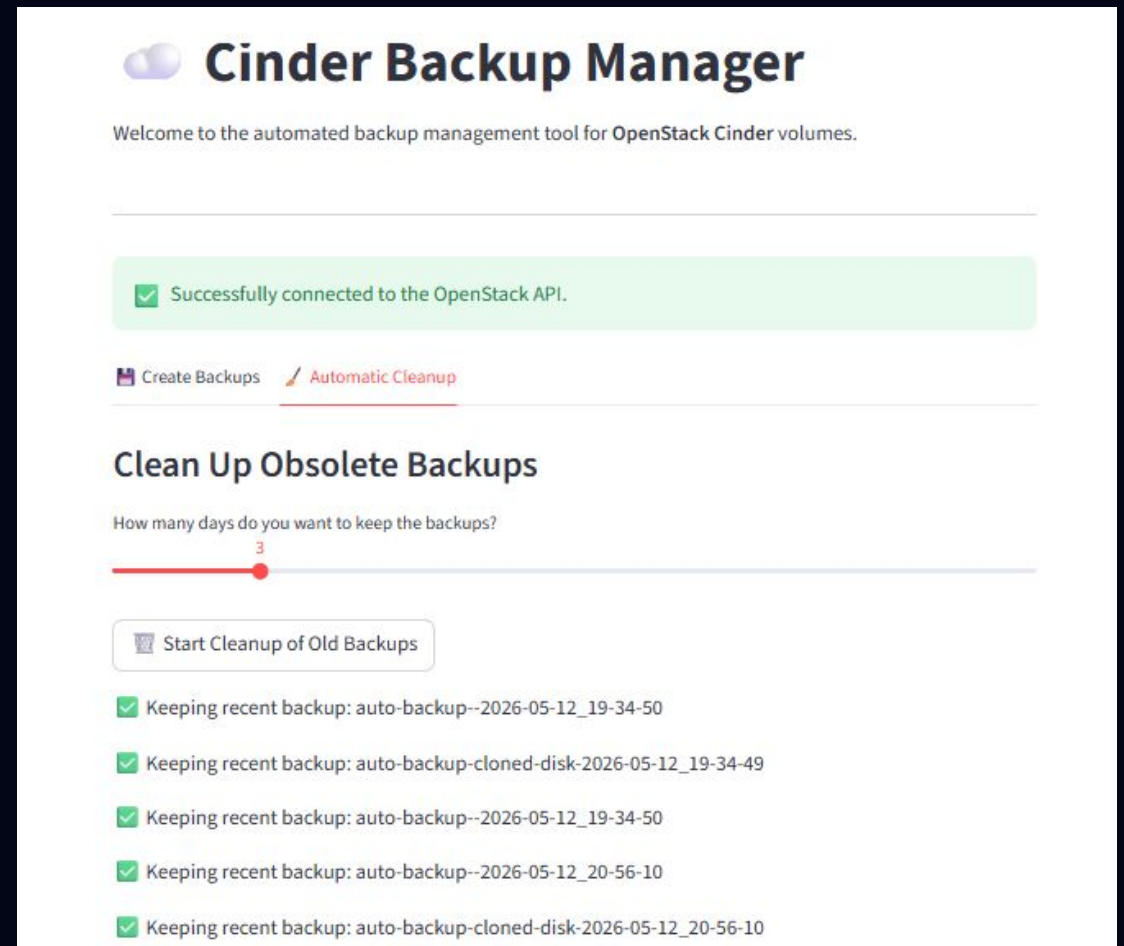
SDK Implementation: Leverages native OpenStack Python bindings for secure, direct communication.

The Cleanup Engine

Technical Challenge: Handling timezone inconsistencies and missing metadata from the API.

The Solution: A multi-layer date parsing strategy:

- Primary: API created_at metadata.
- Fallback: Regex extraction from snapshot names.



Cloud Component: Docker

Deployment: Fully containerized as a stateless microservice.

Security First: Credential injection via .env files. No secrets stored in the image.

Optimization: .dockerignore usage to ensure minimal build size and secure contexts.



Live Demo & Questions

Let's see the Backup Manager in action.

